

DATA PROCESSING & COMPUTER ORGANIZATION (Chapter -1) PART -1

- 1) What is Oldest Counting Machine? → **Abacus.**
- 2) What is the 1st device to help multiplication & division → **Napiers Machine.**
- 3) Pascal's Adding Machine is Called → **Pascaline.**
- 4) Who designed Mechanical Calculator for the 1st time? → **Blaise Pascal.**
- 5) Who is designed an improved Mechanical Calculator? → **Leibnitz.**
- 6) Who is father of Modern Digital Computers? → **Charles Babbage.**
- 7) What was the 1st invention Of Babbage ? → **Difference Engine.**
- 8) What is Analytic Engine?

Ans - The machine designed in a way so that user could feed in a sequence of instructions & data on which to work. The machine could then store & execute the sequence of instructions. But his design featured certain modern concepts like storage capacity, Typewriter for an input device, Punched cards for storing programs etc.

- 9) What is Tabulating Machine?

Ans - **Herman Hollerith** could automatically read the Census data & punched onto cards. The data was coded by a combinations of punched holes on the card.

- 10) Which company 1st develop Tabulating Machine Company?

Ans - **International Business Machines Corporation(IBM)** was 1st develop Tabulating Machine Company.

- 11) What is a Program?

Ans - A Program is a set of instruction indicating to the computer the exact sequence of steps that must be followed to process a given set of data.

- 12) What is ENIAC? What is the use of it?

Ans - The full form of ENIAC is **Electronic Numeric Integrator & Computer.**

It is basically used to calculating **Decimal Nos** instead of **Binary Nos.**

- 13) What is the practical purpose of ENIAC?

Ans - It is used in **scientific research**, calculating range tables for missiles and weather prediction.

- 14) What is the CONS of ENIAC or What is the disadvantages of ENIAC?

Ans - Before it could perform a task, it sometimes take hours to set up the wiring and the switches for a program that would execute in seconds.

- 15) How to overcome the disadvantages of ENIAC?

Ans - **EDVAC(Electronic Discrete Variable Automatic Computer)** is the new circuit to overcome the problem.

EDVAC provide more internal memory than any other existing computer device to be stopped at any point & then again resume work.

- 16) What is **Von Neumann Architecture** of a modern computer?

Ans - Scientist Von Neumann designed a "**Five Function computer**" and of the stored program. This concept is known as Von Neumann Architecture.

- 17) What are the **five Functions** in **Von Neumann Architecture**?

Ans - Memory, Processing, Control, Input and Output are the five functions in Von Neumann Architecture.

- 18) What are the **five components** of **Von Neumann Architecture**?

Ans- i) An Input Unit ii) Arithmetic Logic Unit(ALU) iii) Control Unit(CU) iv) Memory v) Output Unit

- 19) What is the 1st stored program electronic computer?

Ans - **EDSAC(Electronic Delay Storage Automatic Calculator).**

- 20) Comparison between a Calculator & Computer?

Ans - **Similarities -**

- i) Both need **electrical Power** to work.
- ii) Both have an **input device** like a keypad or keyboard.
- iii) Both have a **screen** to display the results.
- iv) Both have **memory** to store data & the results.
- v) Both can do **numerical calculations.**

Dissimilarities -

- i) A computer can do bigger and **more complex** calculations than a calculator.
- ii) A computer has more memory than a calculator.
- iii) Calculator can do only computational job but **computer can do more than computational job.**
- iv) Computer can do **programming in various languages** but calculator has no such facilities.

- 21) **Characteristics of All Generation Computers :-**

Generation	Time	Characteristics
1 st	1942 - 1955	❖ Vacuum Tubes used to build in these generations of computers e.g.- ENIAC. ❖ Very Large in Size. ❖ Very High Power Consumption is required. ❖ Slow processing speed in milliseconds. ❖ Programmed in Machine Language. Eg- UNIVAC-1.
2 nd	1956 - 1964	❖ Transistors were used to build these computers. ❖ These were Smaller, Cheaper, faster more reliable & energy-efficient than 1st Generation Computers. ❖ Commerciably available for business & science. ❖ Magnetic core based Memories used instead of Vacuum tubes. ❖ Machine - independent High Level Programming Languages like COBOL, FORTRAN, etc.

		❖ Floating Point Operations were used. ❖ Processing speeds in Microseconds. Eg - IBM 1401 & IBM 1620.
3rd	1965 - 1970	❖ Integrated Circuits were used instead of Transistors to build these computers. ❖ Smaller, Cheaper, Faster, more reliable & energy-efficient than 2nd Generation Computers. ❖ Semiconductor Memories used instead of Core Memories. ❖ Direct Access Storage Devices(DASD) or Disk Drives were introduced. ❖ Operating System allowed machines to run many different programs in a central machine. ❖ Mass Production of small low cost computers called Minicomputers. Eg - IBM'S System - 360, UNIVAC 1108, RCA 3301, GE 645, HONEYWELL 200.
4th	1971- 1985	❖ Processing Speeds in Nanoseconds. ❖ Microprocessor Technology used to build these computers made them smallest in size. ❖ Cheapest & Fastest among all 3 generations. ❖ Here User Friendly Software Packages are used. ❖ As smaller computers became more powerful, they could be linked together or networked like concept of LAN & WAN developed. Eg- Intel 8085, 8086, 80286, 386, 486, Pentium etc.
5th	1985 to Now	❖ Concept of Parallel processing & Neural Networks. ❖ Use PROLOG(Programming in LOGIC language) for programming. ❖ Computers to process Basic Intelligence to make decisions. ❖ Capable of being a Knowledge Information Processing System(KIPS).

22) What is SSI?

Ans - Small Scale Integration(SSI) contains less than 12 integrated components.

23) What is MSI?

Ans - Medium Scale Integration(MSI) contains about 12 to 100 integrated components.

24) What is LSI?

Ans - Large Scale Integration(LSI) contains more than 100 integrated components per chip.

25) What is VLSI?

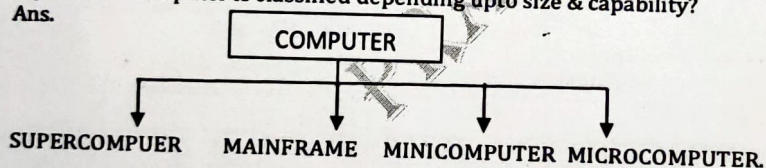
Ans - Very Large Scale Integration(VLSI) contains upto hundreds of thousands of components per chip.

26) What is ULSI?

Ans - Ultra Large Scale Integration(ULSI) contains more than 1 million components on a single chip.

27) How computer is classified depending upto size & capability?

Ans.



SUPERCOMPUTER -

- Among all the supercomputers it is most powerful & most expensive.
- Processing speed is fastest among all computers.
- Data storage capability is highest.
- More than 180 millions data can be calculated per second.
- It's used Parallel Processing technique.
- The price of supercomputers can be more than 100 crores.

Uses - Scientific research, calculating range tables for missiles, Remote Sensing, Medicine Production and weather prediction.

Example - CRAY - 1, CRAY -2, PARAM -2000.

MAINFRAME -

- Size is less than Supercomputer.
- Accessing & Processing speed is less than Supercomputer.
- Storage capacity is huge.
- More than hundred people can work simultaneously.
- Speed is more than min computer & It can calculate more than few crore instruction at a time.

Uses - It is used in Banking Transaction, Bima Sector, Railways, Airways & Hospitals.

Example - IBM S 390, Admahl, Hitachi etc brand Computer.

MINICOMPUTER -

- Here instead of Microprocessor IC or Integrated Circuit were used.
- These computers are bigger in size than Microprocessor, But smaller in size than Mainframe computer.
- These computers are access speed & processing speed & capacity is better than Microprocessor.

- Here simultaneously multiple user can work together.
- These computers are much expensive than Microcomputer but also work better than Microcomputer.

Uses - Financial Accounting work, Networking & DBMS related work, Engineering design like CAD, CAM work, Laboratory research, Institution it is used.

Example - NOVA, DEC, HP3000, NORD 100, SSD 92.

MICROCOMPUTER -

- Microprocessor chips technology were used.
- Size is smaller, Access speed is slow, Storage Capacity less & processing speed is less.
- Cost is least also known as Personal Computer or PC.

Uses - Data storage, Bill generation, Banking Transaction system, Share market these computer are used.

Example - Desktop(PC), Palmtop, Laptop, Notebook etc.

28) Write short notes on Analogue Computer & Digital Computer ?

- **Analogue Computer** - Is usually measured variation in a physical quantity(like voltage, current, pressure etc). Like Thermometer is used to measure the Temperature, Speedometer is used to measure the speed of car, Pressure Gauge(Sphygmomanometer) is used to measure the Blood Pressure, Voltmeter is used to measure the voltage variation.
- **Digital Computer** - These computers are controlled by digital signal. These coding are written in binary 0 or 1 instruction. With this computer input devices like (Keyboard, Mouse) & Output Devices like (Monitor, Printer) is used to control the instruction & process the instruction.

29) Difference between Analogue & Digital Computer ?

Analogue Computer	Digital Computer
1. Calculation process is faster.	1. Calculation process is slower.
2. Less Costly.	2. More costly.
3. These computers are used to calculate a particular task.	3. Multiple task can be calculate a time.
4. Computation are performed as variations of diff. physical quantities.	4. Computations are performed using binary nos.
5. ex- Thermometer, Voltmeter etc	5. ex-microcomputer, Minicomputer, Mainframe Computer & Super computer.

30) Short notes on Von Neumann architecture ?

- In the 1940s, a mathematician called John Von Neumann described the basic arrangement (or architecture) of a computer. Most computers today follow the concept that he described although there are other types of architecture. When we talk about the Von Neumann architecture, we are actually talking about the relationship between the hardware that makes up a Von Neumann-based computer.

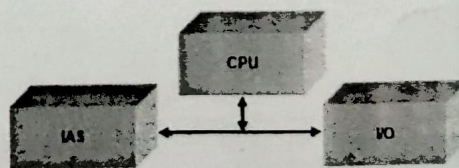
A Von Neumann-based computer is a computer that:

- Uses a single processor.
- Uses one memory for both instructions and data. A von Neumann computer cannot distinguish between data and instructions in a memory location! It 'knows' only because of the location of a particular bit pattern in RAM.
- Executes programs by doing one instruction after the next in a serial manner using a fetch-decode-execute cycle.

In this chapter, we are going to build upon and refine the ideas introduced in an earlier chapter. You should re-read the relevant chapter on CPUs before you start this one. We have already said that the CPU was made up of 4 important components:

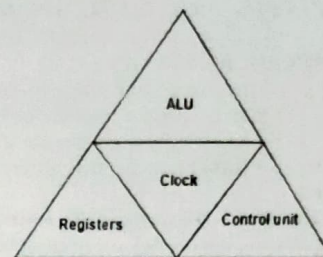
- The ALU.
- The registers.
- The control unit.
- The IAS (otherwise known as RAM or memory).

Because the IAS is so important, we are definitely going to move it to its own section in our model of a computer. (We discussed this previously). We need to get data into and out of the computer so we will include this as a separate section as well. We will also introduce the idea of a clock and clock cycles in the CPU. Our new model of a computer now looks like this:



A model of a Von Neumann computer system.

Von Neumann Component 1 The CPU
The CPU, or Central Processing Unit, is the name given to the component that controls the computer and works on the data. It can be split up into four sub-components:



A diagrammatic representation of the 4 main parts of a CPU.

We know a few things from before about the Von Neumann CPU.

- 1) **The ALU, or Arithmetic Logic Unit** A Von Neumann CPU has an ALU. This is the part of the CPU that performs arithmetic and logic operations on data and acts as the revolving door for the CPU, letting data enter and leave the CPU. We also know that CPUs have a 'word size'. This is the number of bits that can be added, for example, in one go. The bigger a CPU's word size, the more bits it can work on in one clock cycle and the more work you can get done.
- 2) **The Control Unit** A Von Neumann CPU has a control unit. The control unit is in charge of 'fetching' each instruction that needs to be executed in a program by issuing control signals to the hardware. It then decodes the instruction and finally issues more control signals to the hardware to actually execute it.
- 3) **Registers** A Von Neumann CPU has registers. These are very fast memory circuits. They hold information such as the address of the next instruction (Program Counter), the current instruction being executed (Current Instruction Register), the data being worked on and the results of arithmetic and logical operations (Accumulators), information about the last operation (Status Register) and whether an interrupt has happened (Interrupt Register). Registers are covered in a lot more detail later in this chapter.
- 4) **The clock** Instructions are carried out to the beat of the clock! Some instructions take one beat and others more than one beat. Very roughly speaking, the faster the clock, the more clock beats you have per second so the more instructions per second you can do and the faster your computer will go.

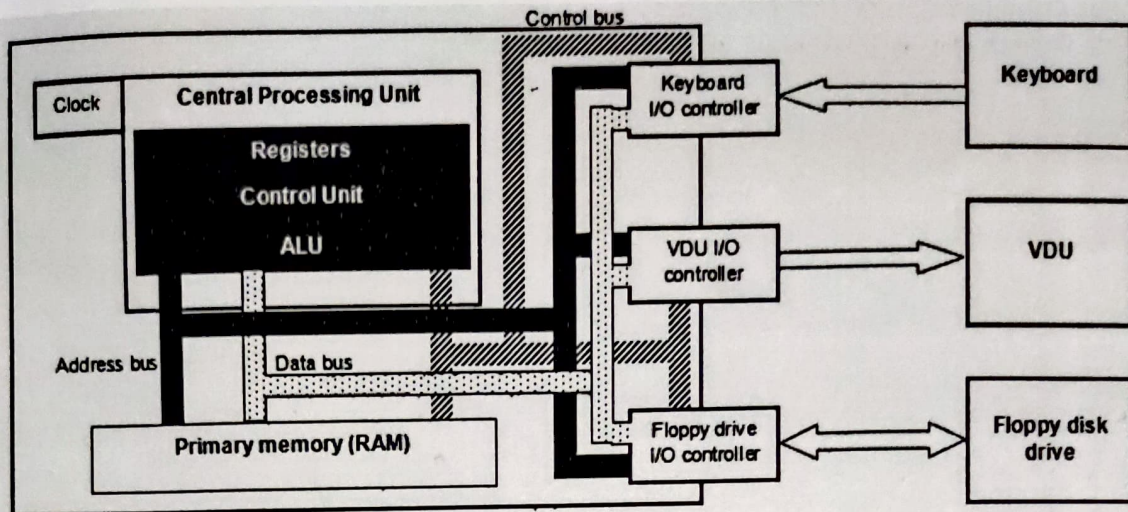
Von Neumann Component 2 - IAS

We also know that the Von Neumann computer has an IAS, or Immediate Access Store, where it puts both programs and data. We often commonly refer to this memory as RAM. RAM is made up of lots of boxes that can store a bit pattern. Each box has a unique address. A memory address might store an instruction (which is made up of an operator and an operand) or it might store just a piece of data. A Von Neumann computer can't tell the difference between the bit patterns as such, but 'knows' indirectly because of where the bit pattern is stored in RAM. Pre-Von Neumann computers used to split up memory into program memory and data memory and this made computers relatively complex. Von Neumann was the first to realise that there was actually no difference between the nature of an instruction and the nature of a piece of data. One important function of an operating system is to manage memory and to keep track of the RAM addresses of applications as well as any data.

We also know that computers have an address bus, so that the CPU can address each individual memory location in the IAS, for example, when it wants to store a piece of data or retrieve a piece of data. The data itself is moved about between devices on a data bus. There is also a control bus, to generate signals to manage the whole process.

Von Neumann Component 3 - I/O

A computer needs peripherals for inputting and outputting data. It needs to be able to read data into itself and send data out. It reads data in and sends data out through its I/O ports. A port is simply a gateway, like a port used for shipping. Just like every port used for ships needs its own harbour master, so every I/O port needs to be managed. An I/O controller is the term used to describe the I/O port along with the circuits that manage data into and out of the port. It allows you to connect up any I/O device to the PC and transfer data in to or out of the computer. You wouldn't want to connect an I/O device directly to a CPU because you would have to redesign the CPU every time a new type of device came along. Besides, a new type of device might need different voltages and control signals from the CPU, again necessitating a CPU redesign. The I/O controller acts as an interface to overcome these problems. Refining the diagram we saw in the previous chapter on CPUs, we now have:



A computer system showing the I/O controllers.

A computer system showing the I/O controllers. Of course, there are a whole range of other I/O controllers we could have included. We could have shown ones for devices such as a mouse, a MIDI device, a printer, a DVD player, a SCSI device as used with many scanners or a network card, to name just a few.

Von Neumann bottleneck Whatever you do to improve performance, you cannot get away from the fact that instructions can only be done one at a time and can only be carried out sequentially. Both of these factors hold back the efficiency of the CPU. This is commonly referred to as the 'Von Neumann bottleneck'. You can provide a Von Neumann processor with more RAM, more cache or faster components but if real gains are to be made in CPU performance then a major review needs to take place of CPU design.

DATA PROCESSING & COMPUTER ORGANIZATION (Chapter -1) PART -2

1) What is the difference between Data & Information ?

Ans.

DATA	INFORMATION
i) Represents raw materials on which computers works.	i) Represents the processed raw material or data.
ii) Data is the starting of Data Processing.	ii) Information is the end product of Data Processing.
iii) Ex- The value of 15042009 by itself is meaningless & serve as the input raw data for a system.	iii) When processed, the same value can represent a date 15-04-2009 etc.

2) Write short notes on Different Input Devices ?

Ans. A) **Keyboard** - The most common input device is the keyboard which is usually connected to the computer using **PS2 or USB Port**. It resembles a mechanical typewriter where the alphabets are placed in the **QWERTY format** along with the numeric 0 - 9 & punctuation symbols.

B) **Mouse** - Mouse is also in an input devices which is used to point & click input device & select any item. It consists of one or more buttons which can be clicked. Mouse can be **wired or Wireless**. Wired mouse generally connected **PS2 or USB Port**. **Wireless mouse use Bluetooth technology to communicate with the computer.**

MOUSE

↓
Mechanical Mouse Optical Mouse

Mechanical Mouse - uses a ball & roller arrangement to determine the position of the mouse pointer,

Optical Mouse - uses an LED light source & a sensor to scan the surface where the mouse is placed & determine the position of the mouse pointer on the screen.

Note - Modern Optical Mouse use invisible IR Laser light as the light source. These are more accurate than the LED mouse & can be used over any surface.

C) **Scanner** - is an input device which can be used to input images & documents & store them in a digital format.

SCANNER

↓
Flatbed Scanner Sheet fed Scanner Handheld Scanner Drum Scanner

i) **Flatbed Scanner** - In these types of Scanner the document is placed on a glass bed underneath which scanner head moves as it scans. The scan head moved with a rotating belt.

ii) **Sheet fed Scanner** - These Scanner have a immobile scan head. Paper document is fed into a roller which rolls the paper in & makes the whole document travel under the fixed scanner head.

iii) **Handheld Scanner** - Similar to Flatbed Scanner but the scan head needs to be moved manually on the document in order to scan. This type of scanner typically does not provide good image quality. But it can be useful to capture text or image quickly.

iv) **Drum Scanner** - These can scan highly detailed images. These use the photomultiplier tube to capture the image. The document to be scanned is mounted on a glass cylinder. A sensor in the centre of the cylinder splits light which is reflected from the document into 3 beams. Each beam is sent through a color filter to a sensor, where the light is changed into an electrical signal.

Note :- Flatbed Scanner is the most widely Used Scanner.

D) **BAR CODE READER** - is used to scan ready product information from product labels. It is used in Supermarkets, Shops, Libraries, Post Office etc.

A standard barcode consists of a number of parallel light & dark bars of variable width. The code is read optically using a special scanner. The barcode is designed based on a 10- digit number system. Of these, 5 digits are used to encode information to the manufacturer & the remaining 5 digits are used to encode the particular product details. The code numbers are also written under each bar symbols which can be used in the event of machine failure.

There are 2 types of Barcode - i) **Uniform Product Code(UPC)** used in US ii) **European Article Number(EAN)** used in Europe.

E) **TOUCH SCREEN** - is a special display screen where the user can enter data by touching the screen at specific locations. It is used in applications like **Personal Digital Assistant(PDA), Mobile Phones, Railway & Airlines Enquiry System, Bank ATMs** etc.

It can directly interact with what is displayed on the screen rather than indirectly with the mouse. **Resistive Touch Screen** is presently the most widely used & affordable touch screen technology which is not affected by heat, dust or water.

F) **LIGHT PEN** - is a point & draw input devices which allows user to point to point objects displayed on a CRT Screen. These are useful where user can directly draw on the screen with the help of a pen. It is similar to Touch Screen but with greater accuracy.

It is basically a light sensitive device that uses a photoelectric cell & an optical lens mounted in a pen shaped case.

G) OPTICAL MARK READER(OMR) - also known as Optical Mark Recognition is the process of detecting data from pre printed document forms marked by humans. Eg. - OMR Answersheet in multiple choice Question examinations.

In these type of sheet an input information by darkening circles marked on the pre printed sheet using HB pencil. The sheet is then automatically read by scanning machine which shines a beam of light onto the OMR form. The contrasting reflective from the marked & unmarked areas is then used to detect the marked areas as they reflect less light than the blank areas of the paper.

But disadvantage is that OMR is not suitable for gathering large amounts of text.

H) OPTICAL CHARACTER READER(OCR) - is basically used to optically scan pages containing text & then to identify and save the scanned data as a text file instead of image file. The image is stored in the computer's memory as a bitmap image consisting of a grid of dots. Then the scanner converts the dots to equivalent ASCII text, which the computer interprets as letters, numbers & special characters.

I) MAGNETIC INK CHARACTER READER (MICR) - is used mainly by Banking system for faster processing of large volumes of cheques. Banks that use MICR use a special type of cheque. The cheque consist of Identification Code, Cheque No., Account No. are preprinted using a special magnetic ink that contains particles of Iron Oxide. E-138 font is used to print the letters in the cheque.

MICR reader is used to recognise the magnetic ink characters automatically and carry out the transaction.

J) WEB CAMERA - used for video capturing & are connected to a computer through an USB Port. Presently computer hardware manufacturers are manufacturing laptops with built in web cams.

K) PUNCHED CARD READERS - The punching machine contains a hopper where the blank cards are 1st stacked, the operator then uses a keyboard to enter the data which is then punched into the card by proper electromechanical mechanisms. To read data from the card, the card is moved over a row of 80 read brushes. If a hole is present a contact is made by the brush. This produces a signal than can be used by the computer to interpret a '0' or a '1'.

L) PUNCHED TAPE READERS- input information punched on paper tape. Here is also a electronic mechanical devices where mechanical sensing pins are used to read the character punched in each line of the tape. The presence & absence of a punch or a hole in the tape is used to encode a binary '0' or a '1'.

M) JOY STICK - is mainly used in computer games & simulation software to input a move. It has special buttons to input various actions required in computer game.

N) MICRO PHONES - can be used to directly input audio signal like voice or music into the computer with the help of a specific software.

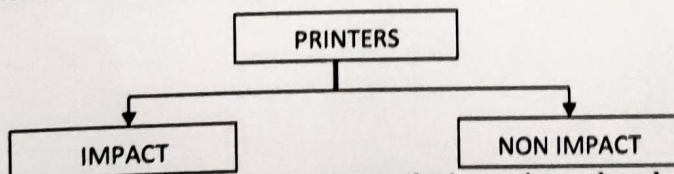
3) What is the difference between OCR & MICR ?

OCR	MICR
Used any type of printed text materials.	Used in banks for reading special information from cheques.
The fonts scanned usually can be variety of shapes.	The fonts scanned should be of a pre defined shape like the E-138 font.
The ink in the text document scanned can be of any type.	The ink used in MICR is magnetic ink made using particles of iron oxide.
The scanned document is usually stored as a text file for editing or Printing.	Information from the scanned data is used for processing cheques in bank.

4) Write short notes on Different Output Devices ?

Ans. A) **VIDEO DISPLAY UNITS (VDU)** - is a video monitor is the most widely used output devices. It is made of a display screen & electronic circuitry that displays a picture corresponding to the video signal sent by the computer. There are 2 main technologies used for manufacturing VDU, a) CRT & LCD display.

B) **PRINTERS** - is used to produce a permanent copy of the output.



IMPACT PRINTERS - In an Impact printers, the image is produced on paper by some electromechanical impact device.

There are various types of Impact Printers like Dot Matrix Printer, Line Printer, Daisy Wheel Printer etc.

i) **Dot Matrix Printer(DMP)** - These printers are used a group of pins of very small radius to strike the ink ribbon & create a dot at the point of impact. A Character of image can be formed by proper arrangement of many such dots.

ii) **Daisy Wheel Printer** - These are character printers where a special type of wheel called Daisy Wheel contains the set of characters used in the printing process. The character are embossed on small metal plates that are mounted on the spokes of a wheel. These printers are fast & good for printing basic text. It can print at the rate of 10 to 50 character per second.

Line Printer - These type of printer prints a complete line of text at a time at very high speed. These printers are much faster than Daisy Wheel Printer but Costly and heavy. Line Printers are of various type -

a) **Drum Printer** - contains cylindrical drum whose length is same as the length of a printed line. The 64 character set is the most common one.

b) **Chain Printer** - The printing characters are chained together to form a loop. This printer can print from 5 to 40 lines per second.

NON- IMPACT PRINTERS- the image is produced on paper without the need for any ink ribbon. The quality of printing is far better & much faster than impact printer. 2 main non impact printers are there -

A) **INKJET PRINTER** - It prints by ejecting microscopic drops of inks through tiny ink nozzles in the print head. Each nozzle can be heated up selectively in a few microseconds using resistive heating effect. On heating the ink droplet in the ink-tank of the ink cartridge expands and is ejected out from the nozzle onto the paper. This ink drop forms a tiny dot on the paper. Several such dots combine to produce a particular character or image. A colour inkjet printer usually contains 4 different ink cartridges or ink tanks - Cyan, Magenta, Yellow & Black(CMYK). Inkjet printers are slower than Laser Printers.

B) **LASER PRINTER** - This type of printers use **DRY INK** called **toner** to print text or images on paper. A laser beam is used to change portions of a roller drum coated with a special photo conducting material. The charge pattern laid down on the drum depends on the content of the document to be printed. Laser printers are costly but can print at high speed of 8 to 15 pages per minute and are suitable for bulk printing.

C) **PLOTTER** - Is similar to an Inkjet Printer but is used for printing on larger papers or continuous paper roll. These are used for printing engineering drawing like building plans, Mechanical Drawings etc.

5) Write down the comparison between VDU & Printer?

Ans.

VDU	PRINTER
Output Device which can display text , graphics or Both	Output Device which can print text,graphics or both.
A VDU can either be in black & white or in colour	A printer can also be in black & white or in colour
Colour VDUs have 3 diff. Assemblies for the primary colours red, green & blue.	A colour printer has four diff. Colour cartridge Cyan, Magenta, Yellow & Black.
The output is temporary & is lost once the computer is switched off.	As the output is usually on paper , it is permanent & can be used later.
No other accessories are required to run a VDU.	Accessories like paper, ink ribbon or ink cartridge are required.
VDU is not provided any memory.	Printers have their own buffer memory.
Eg. - CRT & LCD.	Eg. - DOT MATRIX,INKJET ,LASER PRINTER ETC.

6) What is the difference between Impact & Non Impact Printers ?

Ans. -

IMPACT PRINTERS	NON IMPACT PRINTERS
Print head is operated electromechanically.	Print head does not have an electromechanical device.
Can be used to produce multiple copies by using carbon papers.	Cannot be used to produce multiple copies.
Generally produce character output.	Can produce both character & images.
Usually produce output of a single colour	Can produce output of diff. Colour.
As these are electromechanical devices so they are noisy.	Much less noisy.
Slower than non impact printers.	Faster than Impact Printers.
Eg.- Dot Matrix, Daisy Wheel Printer etc.	Eg. - Inkjet Printer , Laser Printer etc.

7) What is the difference between CRT Monitor & LCD Monitor?

Ans.

CRT Monitor	LCD Monitor
Uses Cathode Ray Tube to generate an Image.	Used Liquid Crystals to generate an image.
Power Consumption is high.	Power Consumption is low.
Displays are either rounder or flat.	Displays are always flat.
Brightness in more.	Brightness in less.
Viewing angle is more.	Viewing angle is less.
These displays are bulky in nature and occupy more space.	These displays are thin in nature and occupy less space.
Less expensive than LCD displays.	More expensive than CRT displays.
Cannot be used as displays for laptops.	Can be used as laptop computer displays.

8) What is Memory Unit ?

Ans. - Computer Data & instruction can be stored in special locations of the computer called Memory unit.

9) How many types of Memory Units are there?

Ans. - Depending upon the type of storage, 2 basic type of Memories are there - 1) Primary Memory 2) Secondary Memory.

a) **Primary Memory** - It is the main memory of the computer & computer cannot run without it. There are 2 types of Primary Memories are there - RAM(Random Access Memory) and ROM(Read Only Memory)

RAM is volatile Memory i.e. used to temporarily store instruction and data that are needed during program execution.

But when the computer is power off automatically the RAM memory will be erased.

ROM is a non volatile Memory that stores instruction during computer start up. The computer reads the instructions permanently stored in the ROM every time power is on.

b) **Secondary Memory** - store large volume data permanently. Eg.- CD,DVD,Pen Drive, External Hard Disk etc.

10) What is Diff. Between Primary Memory and Secondary Memory ?

Ans.

PRIMARY MEMORY	SECONDARY MEMORY
Volatile Memory.	Non Volatile Memory.
Data is lost when power is off.	Data is permanently stored even if power is off.
It is working memory of Computer	It is Storage of Data in the computer.
A computer cannot run without primary memory. Eg. - RAM & ROM.	A computer can run without secondary memory. Eg. - HDD, CD- ROM, DVD, Pen Drive etc.

11) Write Short Notes on ROM?

Ans. ROM is known as Read Only Memory, is a non volatile permanent memory, the content can only be read but cannot be changed. Each computer has the ROM BIOS, which permanently stores the basic information needed by the computer during start up after power is switched on. ROM is also called Firmware.

ROMs are of various type:-

ROM :- These are the 1st Semiconductor ROMs that contained a pre-programmed set of data or instructions. **PROM :-** Programmable ROM. Special equipment called a device programmer is used to write the data in the PROM. The data is written by applying an electrical charge to the input pins of the chip. Once a PROM has been programmed in this way, its content are permanent.

EPROM :- Erasable and Programmable ROM. It works same manner as PROM. But EPROM can be erased & reprogrammed repeatedly. To erase an EPROM it is exposed to a strong source of ultraviolet light. It is sometimes called UVEPROM.

EEPROM :- Electrically Erasable and Programmable ROM. Similar to EPROMs, but instead of UV light the erase operation is done electrically.

TYPE	WRITABLE	COST/BYTE	SPEED
ROM	No	Inexpensive	Fast
PROM	Only Once	Moderate	Fast
EPROM	Many Times	Moderate	Fast
EEPROM	Many Times	Expensive	Fast to read, slow to erase/write.

COMPARISON CHART FOR ROMs

12) Write Short Notes on RAM ?

Ans. Programs are loaded into the main memory & run from there. Data is also loaded into the memory for faster access is called RAM. It is so called because the amount of time required to access data stored in the RAM is the same, independent of the location of the data in the RAM. RAM is volatile in nature & requires a regular supply of power to refresh the contents.

RAM can be classified into 2 :- 1) Static RAM & 2) Dynamic RAM.

DRAM - is a volatile memory that usually consists of a transistor & a capacitor and electrical charge needs to be periodically refreshed to retain the data. As the memory cell needs to be recharged at regular intervals of times, such a RAM is called a dynamic RAM.

SRAM - it can retain the content or data as long as electrical power is applied to the chip. Like DRAM if power is switched off the data will be lost forever. Since these RAM do not require periodic refreshing of the existing data these are called Static RAM. SRAMs are much faster than Dynamic RAM.

13) Write Down the diff. Between RAM & ROM?

Ans. -

RAM	ROM
Full form is Random Access Memory.	Full form is Read Only Memory.
It is temporary memory(Volatile Memory) where the memory content is lost if power is switched off.	It is Permanent memory(Non Volatile memory) where the data are stored Permanently.
Working memory of the computer.	Used to form the BIOS chip of the computer.
One can both Read & Write data to a RAM.	One can only read data from ROM.

14) Write down the comparison Between DRAM & SRAM?

Ans. -

DRAM	SRAM
Volatile Memory	Also a Volatile Memory.
Memory consists of Transistor & Capacitor.	Several Transistors only.
Need to refreshed at regular intervals of time to be retain its data.	No need to refresh at regular intervals of time to retain its data.
Packing density is high	Packing Density is low.
Cost per bit is low.	Cost per bit is high.
Speed 80 nanoseconds.	Speed 20 nanoseconds.
Used to make Main working Memory of Computer.	Used to make cache Memory of a Computer.

15) Write short notes on Cache Memory?

Ans. - Cache memory is also a volatile memory like RAM but much faster than RAM. It is used SRAM technology & hence access time of cache memory is faster than RAM. It functions in between CPU & the Main Memory. The instruction & data i.e. needed frequently is stored in the cache memory of computer for faster access. A computer system can contain more than one cache memory. Like Pentium processors have 2 diff. Cache L1 & L2 Cache. L1 cache is embedded into the processor & L2 Cache is located outside the Processor Chip.

16) Write short notes on Buffer Memory ?

Ans. Buffer memory means memory that is used for temporary storage of data, Buffers are used when there is a diff. Between the rate of data received & the rate at which it can be processed. Buffers can be created in a fixed memory location in hardware. But Majority of buffers are implemented in software, which are typically faster than hardware to store temporary data.

Eg. - When we are printing more than one page through printer at that time printer will store the page data in buffer memory & after that print the data one by one.

17) Write Short notes on Memory Register(MR) ?

Ans. Processor contains several special high speed memory registers located inside the Control Unit(CU) & the CPU. These are used to store the current instruction & data, address of the next instruction to execute and the immediate result of any calculation done by ALU. Register can be 8,16,32,64 bits in size. Sometimes called the Word size. The processor directly communicates with these memory registers to access data & instruction. Hence if word size is more then data can be processed faster by the CPU.

18) Some name & Functions of the Memory Registers ?

Ans.-

NAME	FUNCTIONS
MEMORY ADDRESS REGISTER(MAR)	USED TO HOLD ADDRESS OF CURRENT MEMORY LOCATION.
INSTRUCTION REGISTER(IR)	HOLDS THE CURRENT INSTRUCTION THAT IS BEING EXECUTED.
PROGRAM COUNTER(PC)	HOLDS THE ADDRESS OF NEXT INSTRUCTION TO BE EXECUTED
INPUT/OUTPUT REGISTER(I/O R)	COMMUNICATES WITH THE INPUT/OUTPUT DEVICES.
ACCUMULATOR(ACC)	STORE THE RESULT OF ANY ARITHMETIC OPERATION IN THE ALU.
MEMORY BUFFER REGISTER(MBR)	BUFFER REGISTER USED BY THE CPU TO STORE DATA WHICH IS BEING TRANSFERRED TO OR FROM AN IMMEDIATE STORAGE.

19) Depending upon the way data enters & leaves a register are how many types ?

Ans. - 4 types of Register are there Depending upon the way data enters & leaves a register :-

- Serial In Serial Out.
- Serial in Parallel Out.
- Parallel in Serial Out.
- Parallel in Parallel Out.

20) Write Short note on CPU components ?

Ans. - CPU is the brain of any computer system & made up of 2 basic components :- CU(Control Unit) & ALU(Arithmetic and Logic Unit).

Each Processor has its own set of instruction. The execution of each instruction is done 3 steps :- **Fetch , Decode & Execute operation.**

The 'fetch' operation is used to load the instruction & data into the main memory of the computer.

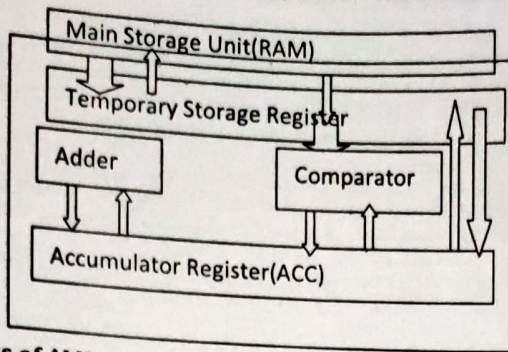
The 'decode' operation is then used to decode the instruction.

The 'execute' operation is finally used to follow the decoded instruction & process the data accordingly.

CONTROL UNIT(CU)-

Functions -

- Control Supervision** -The CU acts as a supervisor unit in the CPU & is responsible for issuing control instructions to carry out diff. Functions of the CPU.
 - Activate Memory Locations**- To read the data & instruction from the main memory , the CU issues control signals to activate memory location . Data & instructions are then transferred from the main memory to special purpose register in the CU.
 - Decode Instruction** - The decode present inside the CU is used to decode an instruction & send appropriate signals to the ALU to process data.
 - Generate Write Signal** - The processed data needs to be transferred from the Temporary memory registers in the CU to the main Memory. The CU issues a write signal to write the processed data to diff. memory locations in the main memory.
 - Control Coordination** - CU is responsible for coordination between the memory and diff. I/O devices.
- The Arithmetic and Logic Unit(ALU)** - is the place where the actual execution of the instruction takes place during data processing. It is responsible for carrying out arithmetic operations like addition, subtraction, multiplication, division etc. And diff. logical operation like >, <, AND, OR etc.



Functions of ALU :-

- i) Fetch Data.
- ii) Carry Out Calculation.
- iii) Compare Data.
- iv) Logical Operation.
- v) Store Processed Data.

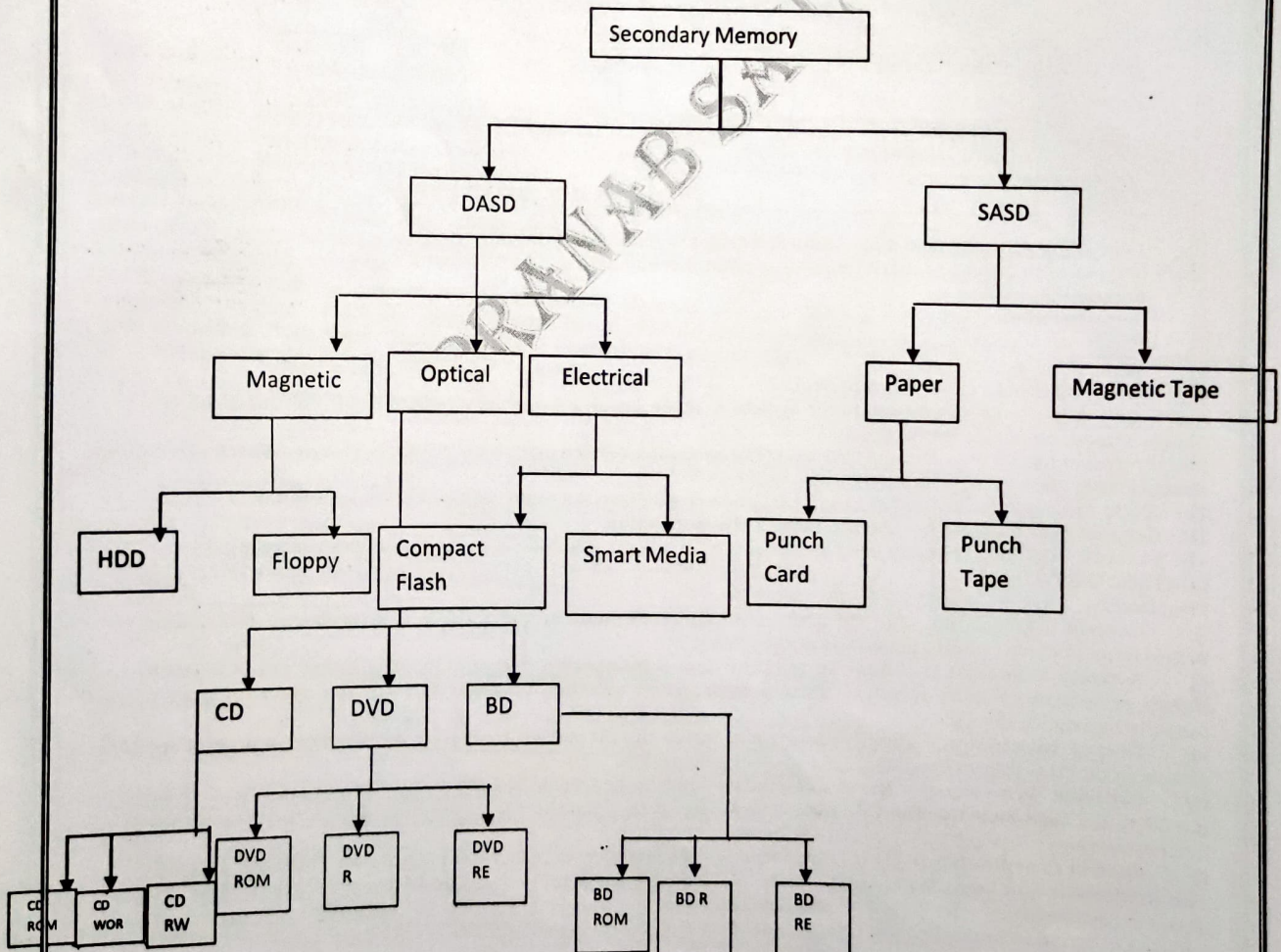
21) What is MicroProcessor ? Write short notes on Microprocessor?

Ans. When the function of a CPU are integrated on a single integrated circuit(IC) using Ultra Large Scale Integration(ULSI) technology then it is called a Microprocessor.

In microprocessor trillions of Transistors that are used to form the diff. components of a CPU are packed into a single chip.

Eg.- At 1st 4 bit microprocessor were used then 8 bit, 16 bit, 32 bit and now recent 64 bit microprocessors are used in AMD Athlon's X2, Intel's Pentium D, Core 2 Duo, Core i3, i5, i7 processors.

22) Write the tree view of Secondary Memory or Secondary Storage Devices



23) Write Short notes on all Secondary Memory?

Ans.

Name of the Secondary Drive	Function
Magnetic Hard Disks	It is a Direct Access Storage Devices(DASD) used to store data digitally. In a hard disk data is stored permanently on 1 or multiple circular disks called platters. To read or write data onto the disk, a motor spins the disk at a very high speed of around 7200 rounds per minute.
Magnetic Floppy Disks	It is also a Direct Access Storage Devices(DASD) floppy disk which is a removable plastic diskette & has a much lower storage capacity. Floppy disk capacity ranges between 1.2 MB - 1.44 MB.
Magnetic Tape Media	Is a sequential Access Storage Device(SASD) , drive can move its read/write heads to any random part of the disk platters in a very short amount of time. The data density is in Bytes per inch(BPI) or Characters per inch(CPI). The Tape speed expressed in inches per second(IPS) .
Compact Disk(CD)	CD is a DASD made of polycarbonate plastic disk. CD capacity is 700 MB
	CD-ROM is used to read data from CD. CD-ROM is a read only media.
	CD-R is a CD- Recordable Media, also called a Write Once Read Many(WORM) type media. Where user can write data on the CD only once but read it many times.
Digital Versatile Disk(DVD)	CD-R/W is a CD Re-Writable Media, Which can both read & written many times.
	Is an Optical Storage Media like a CD but stores much more data than CD. Maximum Capacity of a Dual layer DVD is about 8.54 GB as compared to CD of 700 MB. DVD with single layer can store 4.70 GB data.
Blu ray Disk(BD)	Is an optical disk storage medium can have capacity of upto 50 GB. Its main uses are High definition Video & Data Storage. Single Layer BD also store upto 25 GB of Data & Dual Layer store upto 50 GB of Data.
	BD-ROM is used to read data from BD. BD-ROM is a read only media.
	BD-R Recordable Media, also called a Write Once Read Many(Worm) type media.
	BD-RW is a BD Re-Writable Media, which can both read & written many times.
Flash Memory	Is a solid state storage device which means that there are no mechanical moving parts everything is done electronically. Works similar to EEPROM . Eg. - Smart Media, Compact Flash, Pen Drive
	i) Smart Media - Successor to Floppy Disk. Also called Solid State Floppy Disk Card(SSFDC), Capacity 0.50 MB to 128 MB.
	ii) Compact Flash - largest capacity 32 GB. Transfer rate -45 MB/Sec. Used in Digital Cameras to store High Resolution Digital Images.
	iii) Pen Drive - Flash memory integrated with a USB Port. Capacity from 4 GB - 64 GB. Suitable for transporting data from one place to another.

24) What are the Disadvantages of Magnetic Disk ?

Ans. i) Lower efficiency for sequential processing.

- ii) Low security.
- iii) Prone to easy data loss.
- iv) Higher cost per bit than tape.
- v) Less Portable.
- vi) Needs special care.

25) What is the advantage of Magnetic Disk over Magnetic Tape?

Ans. -

- i) Random Access of Data.
- ii) Fast Data Read/Write.
- iii) Shared Device.
- iv) High Data Trasfer Rate,
- v) More Durable.

26) What is System Bus ?

Ans. System Bus connects the diff. parts of computer together.

System Bus can be divided into 2 parts Internal Bus & External Bus.

System Bus have 3 main components are as follows :-

Name of the Component	Description
1. Address Bus <i>Bi-directional</i>	Connects the CPU & RAM & carries the Memory Addresses. It is necessary to specify the address of the memory location where the data is stored.
2. Data Bus <i>After trasferring data it works as Bi-directional</i> <i>Before trasferring data it works as Uni-directional</i>	Connects the CPU , Memory & other Hardware locations from where data needs to be transferred. A 32 bit bus can transfer 4 bytes of data & 64 bit bus can transfer 8 bytes of data at a time.
3. Control Bus <i>Bi-directional</i>	In addition to sending addresses & exchanging data with the memory , the CPU also needs to send control signals to the memory to specify whether the data is to be read from or written to the specified address location. It carries signals, which are in the form of read/write commands.

27) What is Universal Serial Bus?

Ans. - Using an USB upto 127 devices can be attached to a computer. An USB cable consist of 4 wires - 2 for data, 1 for Power & 1 for ground. Presently USB is the most Popular Communication Bus used to attach to a computer.

4. Power Bus

[Uni-directional]

DATA PROCESSING & COMPUTER ORGANIZATION SHORT QUESTION**Chapter 1 -**

- 1) The Abacus was invented by - **Chinese**.
- 2) In 1617 John Naper of Scotland invented a calculating Device called - **Napier's Machine**.
- 3) In 1642 Blaise Pascal of France invented a calculating machine called - **Pascaline**.
- 4) The Frenchmen named Colmar invented the calculating machine called - **Arithometer**.
- 5) In 1823 a calculating machine called the Difference Engine was developed by - **Charles Babbage**.
- 6) The punched card reader was developed by the American inventor - **Herman Hollerith**.
- 7) The Transistor was invented by - **Shockley**.
- 8) Jack Kilby, an engineer with Texas Instruments developed the - **Integrated Circuit**.
- 9) Computers used mainly in Scientific Research Organizations for processing complex scientific data that require huge processing power are called - **Super Computer**.
- 10) Computers used in large organisations such as banks, railways,airways etc. That need frequent access to the same information - **Mainframes**.
- 11) The 1st IBM PC used the - **8088 μ processor**.
- 12) Multimedia systems require - **Speaker, Scanner, Web Camera**.

Chapter 2 -

- 1) Raw Data is processed to get a more meaningful interpretation of the data is called - **information**.
- 2) The Process of converting data to information is called - **Data Processing**.
- 3) A computer stores single pieces of information in units is called - **Attributes**.
- 4) Data Processing by a computer involves generally 3 basic steps input,Process & - **Output**.
- 5) A sequence of instructions that needs to be given to the computer in a language that the computer understands is called - **Software**.
- 6) In case a part of the software is permanently stored in a computer in a read only type memory then such software is called - **Firmware**.
- 7) In a standard computer keyboard the alphabets are placed in - **QWERTY Order**.
- 8) A mouse is a point & click input device used to point & select an item on a - **GUI**.
- 9) A _____ is an input device which can be used to input images and documents into the computer- **Scanner**.
- 10) A _____ is an input device used to scan ready product information from product labels - **Bar Code Reader**.
- 11) In an impact type printer, the image is produced on a paper by - **Electromechanical Impact**.
- 12) Line Printer is an impact printer used to print - **A Complete Line**.
- 13) In which printer type an image is formed on paper without the need for any hammer or ink ribbon ?
Ans.- **Laser**.
- 14) Two types of Primary Memory are - **RAM & ROM**.
- 15) Secondary Memory is a - **Non Volatile Memory**.
- 16) The term used to refer to an area of the memory that is used for temporary storage of data - **Buffer Memory**.
- 17) Data is recorded on a hard disk using what physical property of a material - **Magnetic**.
- 18) Magnetic hard disk can be used only after they are prepared by a process called - **Disk Formatting**.
- 19) During formatting, a pattern is laid out on the disk which divides the surface of the disk into a number of invisible concentric logical circles called - **Track**.
- 19) Each track is further subdivided into smaller sections called - **Sector**.
- 20) Time required to position the read/Write head from the current track to the desired track is - **Seek Time**.
- 21) Time required to position the desired sector under the read/write head once the track is selected - **Latency**.
- 22) To avoid contamination, the hard disk patterns are nowadays stored in a sealed airtight box using a technology called - **Winchester Technology**.
- 23) Unlike magnetic hard disks, a magnetic tape is - **SASD**.
- 24) The electrical path that connects the CPU and RAM and carries the memory address - **Address Bus**.
- 25) The electrical path that connects the CPU, memory and other hardware locations from where data needs to be transferred - **Data Bus**.